Relativistic Closure–Curvature Bridge Theorem: A Full Long-Form Theorem Construction

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Abstract

A full theorem-level bridge is constructed from the pre-geometric closure framework of Darke Theory to an effective relativistic curvature regime. The construction proceeds from primitive objects, threshold activation laws, an admissibility calculus, constitutive closure transport, a concrete matter sector, and a coupled action to a chain of lemmas and theorems establishing: post-threshold geometric activation, well-posed admissibility structure, hyperbolic causal propagation, Lorentzian regime emergence, matter as persistent topological coherence, gravitational consistency, weak-field reduction, perturbative vacuum stability, a homogeneous closure-cosmology reduction compatible with Darke Theory's non-Friedmann expansion logic, and the final relativistic closure–curvature bridge theorem. The purpose is not to outline a programme, but to state the bridge itself in full mathematical form.

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1 Statement of Purpose

This document is a theorem construction, not a broad conceptual summary and not a short note. Its purpose is to build, in full form, a relativistic bridge from the post-threshold closure architecture of Darke Theory to an effective curvature and matter regime.

The bridge is designed to establish, in one coupled system, the following claims:

1. geometry activates only above a closure threshold;
2. admissibility is represented by a real functional on the field space;
3. curvature is sourced by coherence gradients, matter sectors, and admissibility stress;
4. causal structure arises from hyperbolicity and is physically filtered by closure balance;
5. stable topological coherence sectors act as matter-energy concentrations;
6. the coupled metric–coherence–matter system is gravitationally consistent on-shell;
7. the post-threshold vacuum is linearly stable under explicit positivity conditions;
8. a homogeneous cosmological sector exists without presupposing the standard Friedmann equation as primitive.

The theorem chain below is therefore the bridge itself.

2 Primitive Objects and Post-Threshold Regime

Definition 1 (Primitive manifold)**.** *Let M be a smooth four-manifold.*

Definition 2 (Primitive fields)**.** *Let the primitive post-threshold field content be:*

- *a smooth symmetric nondegenerate tensor $g_{\mu\nu}$;*
- *a coherence-density field $\Phi : M \to \mathbb{R}_{>0}$;*
- *a complex closure-matter field $\psi : M \to \mathbb{C}$.*

Definition 3 (Closure threshold)**.** *Let $\Phi_{\mathrm{crit}} > 0$ be the closure threshold and define the dimensionless closure order parameter*

$$\chi := \frac{\Phi - \Phi_{\mathrm{crit}}}{\Phi_0}, \qquad \Phi_0 > 0.$$

Definition 4 (Post-threshold closure regime)**.** *The post-threshold regime is the field domain on which*

$$\Phi(x) > \Phi_{\text{crit}} \qquad \forall x \in M.$$

Remark 5. *The bridge theorem concerns only the post-threshold regime. No geometric or relativistic claim is made below threshold.*

3 Axioms of the Bridge Construction

Axiom 6 (Threshold activation)**.** *Geometry is activated only in the post-threshold regime.*

Axiom 7 (Finite support)**.** *Persistence of a regime requires finite support and is not granted an unbounded carrying capacity by default.*

Axiom 8 (Admissibility filter)**.** *Only field configurations with nonnegative total admissibility can qualify as persistent regimes.*

Axiom 9 (Closure-sensitive causality)**.** *Physical propagation must lie within a hyperbolic propagation cone and remain compatible with nonnegative closure balance.*

Axiom 10 (Stable matter as closed coherence)**.** *Matter is represented by finite-energy nontrivial topological sectors of a concrete closure-matter field.*

These axioms are the operational translation of the pre-geometric closure architecture into the bridge construction.

4 Geometry Activation Law

Definition 11 (Activation function)**.** *A geometry-activation function is a smooth map*

$$\sigma : \mathbb{R} \to (0, 1)$$

such that

$$\sigma'(\chi) > 0, \qquad \lim_{\chi \to -\infty} \sigma(\chi) = 0, \qquad \lim_{\chi \to +\infty} \sigma(\chi) = 1.$$

Definition 12 (Curvature activation coupling)**.** *Define*

$$F(\Phi) := \sigma(\chi)(1 + \xi\chi + \eta\chi^2),$$

where the parameters are chosen so that

$$F(\Phi) > 0 \qquad \textit{for all } \Phi > \Phi_{\text{crit}}.$$

Lemma 13 (Post-threshold geometric activation)**.** *If $\Phi > \Phi_{\text{crit}}$, then $F(\Phi) > 0$.*

Proof. In the post-threshold regime, $\chi > 0$. By construction, $\sigma(\chi) \in (0, 1)$, and the polynomial prefactor is assumed positive on the post-threshold domain. Hence $F(\Phi) > 0$. □

Remark 14. *This is the precise threshold mechanism by which the geometric sector enters the bridge.*

5 Concrete Matter Sector

Definition 15 (Closure-matter Lagrangian)**.** *Define*

$$\mathcal{L}_{\mathrm{m}} = -\frac{1}{2} K(\Phi)\, g^{\mu\nu} \nabla_\mu \psi\, \nabla_\nu \overline{\psi} - U(\psi, \Phi),$$

where

$$K(\Phi) > 0$$

and

$$U(\psi, \Phi) = \lambda_\psi \left(|\psi|^2 - v(\Phi)^2 \right)^2 .$$

Definition 16 (Broken-phase vacuum manifold)**.** *The broken-phase vacuum manifold is*

$$|\psi| = v(\Phi).$$

Definition 17 (Topological charge)**.** *For a closed loop γ in a nontrivial sector, define*

$$Q[\psi] = \frac{1}{2\pi} \oint_\gamma d(\arg \psi).$$

Definition 18 (Finite-energy topological sector)**.** *A matter configuration is a finite-energy topological sector if*

$$E[\psi] < \infty, \qquad Q[\psi] \neq 0.$$

6 Closure Transport and Closure Balance

Definition 19 (Closure transport current)**.** *Define the constitutive closure current*

$$J^\mu = \zeta(\Phi)\, \nabla^\mu \Phi, \qquad \zeta(\Phi) \geq 0.$$

Definition 20 (Closure-balance scalar)**.** *Define*

$$\Sigma := \nabla_\mu J^\mu.$$

Interpret Σ through the constitutive balance law

$$\Sigma = \mathcal{C}(\Phi, \psi) - \mathcal{D}(\Phi, \psi),$$

where $\mathcal{C} \geq 0$ is closure generation and $\mathcal{D} \geq 0$ is closure loss.

Remark 21. *Hyperbolicity determines the kinematic propagation cone. Closure balance then filters the physically admissible subcone.*

7 Admissibility Calculus

7.1 Burden and Support Densities

Definition 22 (Local burden density)**.** *Define*

$$\mathcal{B} = \alpha_1\, g^{\mu\nu} \nabla_\mu \Phi\, \nabla_\nu \Phi + \alpha_2\, \Sigma^2 + \alpha_3\, K(\Phi)\, g^{\mu\nu} \nabla_\mu \psi\, \nabla_\nu \overline{\psi} + \alpha_4\, U(\psi, \Phi),$$

where $\alpha_i > 0$.

Definition 23 (Total burden)**.** *Define*

$$B[\Phi, \psi; g] = \int_M d^4x\, \sqrt{-g}\, \mathcal{B}.$$

Definition 24 (Local support density)**.** *Define*

$$\mathcal{S} = \beta_1 \Phi - \beta_2 \Phi^{-1} g^{\mu\nu} \nabla_\mu \Phi \, \nabla_\nu \Phi - \beta_3 \Xi[g] - \beta_4 |\psi|^2,$$

where $\beta_i > 0$ *and* $\Xi[g] \geq 0$.

Definition 25 (Total support)**.** *Define*

$$S_{\text{avail}}[\Phi, \psi; g] = \int_M d^4x \, \sqrt{-g} \, \mathcal{S}.$$

Definition 26 (Admissibility functional)**.** *Define*

$$\mathcal{A}[\Phi, \psi; g] := S_{\text{avail}}[\Phi, \psi; g] - B[\Phi, \psi; g].$$

Definition 27 (Admissible configuration)**.** *A field configuration is admissible if*

$$\mathcal{A}[\Phi, \psi; g] \geq 0.$$

7.2 Smooth Inadmissibility Penalty

Definition 28 (Smooth penalty)**.** *Define*

$$\mathcal{P}(\mathcal{A}) = \frac{1}{2} \left(\sqrt{\mathcal{A}^2 + \varepsilon^2} - \mathcal{A} \right)^2, \qquad \varepsilon > 0.$$

Lemma 29 (Penalty properties)**.** *The penalty* $\mathcal{P}$ *is smooth, nonnegative, and satisfies:*

1. $\mathcal{P}(\mathcal{A}) \approx 0$ *when* $\mathcal{A} \gg 0$,
2. $\mathcal{P}(\mathcal{A}) \sim 2\mathcal{A}^2$ *when* $\mathcal{A} \ll 0$.

Proof. Smoothness follows from $\varepsilon > 0$. Nonnegativity follows because the expression is a square. The asymptotics follow by expansion at large positive and negative $\mathcal{A}$. □

7.3 Well-Posedness

Theorem 30 (Admissibility Functional Well-Posedness Theorem)**.** *Assume:*

1. $Z(\Phi) > 0$,
2. $K(\Phi) > 0$,
3. $U(\psi, \Phi) \geq 0$,
4. $\Xi[g] \geq 0$,
5. $\mathcal{C}(\Phi, \psi) \geq 0$,
6. $\mathcal{D}(\Phi, \psi) \geq 0$,
7. $\alpha_i, \beta_i > 0$.

Then:

1. $B[\Phi, \psi; g] \geq 0$,
2. $\mathcal{P}(\mathcal{A}) \geq 0$,
3. *the penalty* $\mathcal{P}$ *is smooth,*
4. *the full action defined later is bounded below by its potential and penalty sectors on the admissible finite-energy class.*

Proof. By the positivity assumptions, each term in $\mathcal{B}$ is nonnegative, so $B \geq 0$. By Theorem 29, the penalty is smooth and nonnegative. The matter and coherence potentials are quartic and bounded below. Therefore the full action is bounded below on the admissible finite-energy class. □

8 Full Bridge Action

Definition 31 (Coherence potential)**.** *Define*

$$V(\Phi) = \Lambda_0 + \frac{\mu^2}{2}(\Phi - \Phi_{\text{crit}})^2 + \frac{\lambda}{4}(\Phi - \Phi_{\text{crit}})^4.$$

Definition 32 (Positive coherence kinetic weight)**.** *Assume*

$$Z(\Phi) > 0.$$

Definition 33 (Bridge action)**.** *The full coupled bridge action is*

$$S[g, \Phi, \psi] = \int_M d^4x \sqrt{-g} \left[\frac{1}{2\kappa} F(\Phi) R - \frac{Z(\Phi)}{2} g^{\mu\nu} \nabla_\mu \Phi \, \nabla_\nu \Phi - V(\Phi) + \mathcal{L}_{\text{m}} - \gamma \, \mathcal{P}(\mathcal{A}) \right].$$

9 Euler–Lagrange System

9.1 Metric Equation

Proposition 34 (Metric field equation)**.** *Variation with respect to $g^{\mu\nu}$ gives*

$$F(\Phi) G_{\mu\nu} = \kappa T^{(\Phi)}_{\mu\nu} + \kappa T^{(\psi)}_{\mu\nu} + \nabla_\mu \nabla_\nu F - g_{\mu\nu} \Box F - \kappa V(\Phi) g_{\mu\nu} - \kappa\gamma \, \Pi^{(\mathcal{A})}_{\mu\nu},$$

where

$$T^{(\Phi)}_{\mu\nu} = Z(\Phi) \nabla_\mu \Phi \nabla_\nu \Phi - \frac{1}{2} Z(\Phi) (\nabla\Phi)^2 g_{\mu\nu},$$

$$T^{(\psi)}_{\mu\nu} = K(\Phi) \nabla_\mu \psi \nabla_\nu \overline{\psi} - g_{\mu\nu} \mathcal{L}_{\text{m}},$$

and

$$\Pi^{(\mathcal{A})}_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta}{\delta g^{\mu\nu}} (\sqrt{-g} \, \mathcal{P}(\mathcal{A})).$$

9.2 Coherence Equation

Proposition 35 (Coherence field equation)**.** *Variation with respect to Φ gives*

$$Z(\Phi) \Box \Phi + \frac{1}{2} Z'(\Phi) (\nabla\Phi)^2 - V'(\Phi) + \frac{1}{2\kappa} F'(\Phi) R + \frac{\partial \mathcal{L}_{\text{m}}}{\partial \Phi} - \gamma \frac{\delta \mathcal{P}(\mathcal{A})}{\delta \Phi} = 0.$$

9.3 Matter Equation

Proposition 36 (Matter field equation)**.** *Variation with respect to $\overline{\psi}$ gives*

$$\nabla_\mu (K(\Phi) \nabla^\mu \psi) - \frac{\partial U}{\partial \overline{\psi}} - \gamma \frac{\delta \mathcal{P}(\mathcal{A})}{\delta \overline{\psi}} = 0.$$

10 Hyperbolic Causal Structure

10.1 Principal Symbol

The principal part of the coherence equation is

$$Z(\Phi) \, g^{\mu\nu} \nabla_\mu \nabla_\nu \Phi.$$

Thus the principal symbol is

$$\sigma_\Phi(\xi) = Z(\Phi) \, g^{\mu\nu} \xi_\mu \xi_\nu.$$

Definition 37 (Admissible causal cone)**.** *At each point $x \in M$, define*

$$\mathcal{C}_x^{\mathrm{adm}} = \{v^\mu \in T_x M : g_{\mu\nu} v^\mu v^\nu \leq 0, \quad v^\mu \nabla_\mu \Sigma \geq -\Sigma_*\},$$

where $\Sigma_* \geq 0$ is a fixed closure-loss threshold.

Theorem 38 (Hyperbolic Causal Closure Theorem)**.** *Assume:*

1. $F(\Phi) > 0$,
2. $Z(\Phi) > 0$,
3. $g_{\mu\nu}$ *is nondegenerate,*
4. *the principal symbol* $g^{\mu\nu}\xi_\mu\xi_\nu$ *is hyperbolic.*

Then the post-threshold regime admits a well-defined causal cone. If, in addition,

$$\Sigma = \mathcal{C}(\Phi, \psi) - \mathcal{D}(\Phi, \psi) \geq 0$$

along physical transport histories, then realised propagation lies inside the admissible causal cone $\mathcal{C}_x^{\mathrm{adm}}$.

Proof. Hyperbolicity of the principal symbol defines the null cone of the coherence operator. Since $Z(\Phi) > 0$, the same cone is determined by $g_{\mu\nu}$. The closure-balance condition excludes directions along which closure renewal fails. Therefore physical propagation lies in the hyperbolic cone refined by closure admissibility. □

11 Lorentzian Regime Emergence

Theorem 39 (Lorentzian Regime Theorem)**.** *Assume:*

1. $F(\Phi) > 0$,
2. $Z(\Phi) > 0$,
3. $g_{\mu\nu}$ *is smooth, symmetric, and nondegenerate,*
4. *the principal symbol of the coherence equation is hyperbolic,*
5. *the quadratic form* $g^{\mu\nu}\xi_\mu\xi_\nu$ *has exactly one negative eigenvalue class at each point.*

Then the emergent geometric regime is Lorentzian.

Proof. A smooth, symmetric, nondegenerate bilinear form with one negative eigenvalue class has Lorentzian signature up to convention. Hyperbolicity gives the distinguished causal cone. Hence the post-threshold regime is Lorentzian. □

12 Matter as Persistent Closed Coherence

Theorem 40 (Matter as Persistent Closed Coherence Theorem)**.** *Let ψ satisfy the matter field equation in the post-threshold regime. Assume:*

1. $K(\Phi) > 0$,
2. $U(\psi, \Phi) \geq 0$,
3. $E[\psi] < \infty$,

4. $Q[\psi] \neq 0$.

Then ψ defines a nontrivial persistent topological coherence sector whose stress-energy $T^{(\psi)}_{\mu\nu}$ acts as a persistent curvature source. Such sectors are macroscopically interpretable as matter-energy concentrations.

Proof. Finite energy implies localisation and admissibility. Nonzero topological charge prevents continuous relaxation to the trivial vacuum sector within the finite-energy class. Hence the configuration persists as a nontrivial topological coherence sector. Since $T^{(\psi)}_{\mu\nu} \neq 0$ on its support, the metric equation identifies it as a persistent curvature source. □

13 Gravitational Consistency

Define

$$T^{\text{eff}}_{\mu\nu} = T^{(\Phi)}_{\mu\nu} + T^{(\psi)}_{\mu\nu} + \kappa^{-1}\left(\nabla_\mu\nabla_\nu F - g_{\mu\nu}\Box F\right) - V(\Phi)g_{\mu\nu} - \gamma\Pi^{(\mathcal{A})}_{\mu\nu}.$$

Then

$$F(\Phi)G_{\mu\nu} = \kappa T^{\text{eff}}_{\mu\nu}.$$

Lemma 41 (Penalty divergence identity)**.** *On-shell,*

$$\nabla^\mu\Pi^{(\mathcal{A})}_{\mu\nu} = \frac{1}{\sqrt{-g}}\frac{\delta\mathcal{P}}{\delta\Phi}\nabla_\nu\Phi + \frac{1}{\sqrt{-g}}\frac{\delta\mathcal{P}}{\delta\psi}\nabla_\nu\psi + \frac{1}{\sqrt{-g}}\frac{\delta\mathcal{P}}{\delta\overline{\psi}}\nabla_\nu\overline{\psi}.$$

Proof. The penalty contribution is a diffeomorphism-invariant scalar density built from $(g_{\mu\nu}, \Phi, \psi, \overline{\psi})$. The identity follows from its Noether variation under an infinitesimal diffeomorphism and the corresponding Euler–Lagrange terms. □

Theorem 42 (Gravitational Consistency Theorem)**.** *Assume the Φ- and ψ-equations hold and the action is diffeomorphism invariant. Then*

$$\nabla^\mu\big(F(\Phi)G_{\mu\nu} - \kappa T^{\text{eff}}_{\mu\nu}\big) = 0.$$

Proof. By the contracted Bianchi identity,

$$\nabla^\mu G_{\mu\nu} = 0.$$

Hence

$$\nabla^\mu(FG_{\mu\nu}) = (\nabla^\mu F)G_{\mu\nu}.$$

The divergence of the $F(\Phi)R$-variation terms cancels this contribution when the coherence equation holds. The divergence of $T^{(\psi)}_{\mu\nu}$ vanishes on-shell by the matter equation. The penalty contribution is handled by Theorem 41. Thus the full equation is covariantly conserved. □

14 Weak-Field Reduction

Let

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad \Phi = \Phi_* + \varphi, \qquad \Phi_* > \Phi_{\text{crit}}, \qquad V'(\Phi_*) = 0.$$

Then

$$F(\Phi) = F_* + F'_*\varphi + O(\varphi^2).$$

Theorem 43 (Weak-Field Closure–Curvature Limit)**.** *In the post-threshold vacuum, the bridge system reduces at first order to linearised relativistic gravity with coherence-scalar and admissibility corrections:*

$$F_*G^{(1)}_{\mu\nu} = \kappa T^{(\psi,1)}_{\mu\nu} + \kappa T^{(\varphi,1)}_{\mu\nu} + \left(\partial_\mu\partial_\nu - \eta_{\mu\nu}\Box\right)\left(F'_*\varphi\right) - \kappa\gamma\Pi^{(\mathcal{A},1)}_{\mu\nu}.$$

Proof. Linearise the metric and coherence equations about the post-threshold stationary vacuum. Retaining first-order terms yields the stated system. □

15 Vacuum Stability

Theorem 44 (Post-Threshold Vacuum Perturbative Stability Theorem)**.** *Assume:*

1. $F(\Phi_*) > 0$,
2. $Z(\Phi_*) > 0$,
3. $V''(\Phi_*) > 0$,
4. $K(\Phi_*) > 0$.

Then the post-threshold vacuum

$$(g_{\mu\nu}, \Phi, \psi) = (\eta_{\mu\nu}, \Phi_*, \psi_*)$$

is linearly stable at first order against small coherence and matter perturbations.

Proof. The linearised coherence mode has positive kinetic coefficient $Z(\Phi_*)$ and positive effective mass $V''(\Phi_*)$, so it is non-ghost and non-tachyonic at first order. The matter sector has positive kinetic coefficient $K(\Phi_*)$, so its linearised propagation is likewise well-signed. The condition $F(\Phi_*) > 0$ preserves the correct sign of the effective gravitational coupling in the weak-field regime. Therefore the background is linearly stable at first order against small coherence and matter perturbations. □

16 Homogeneous Closure-Cosmology Reduction

The bridge theorem must not overwrite the Darke Theory cosmology by silently importing the standard Friedmann equation as primitive. The correct reduction is therefore stated as an *effective homogeneous closure-cosmology sector*, compatible with the existing separation

$$H(t) = H_{\text{rec}}(t) + H_{\text{flow}}(t),$$

rather than as an axiomatic commitment to standard GR cosmology.

16.1 Homogeneous Regime

Assume a homogeneous, isotropic sector with

$$ds^2 = -dt^2 + a(t)^2 d\Sigma_k^2, \qquad \Phi = \Phi(t), \qquad \psi = \psi(t).$$

Define the expansion rate

$$H := \frac{\dot{a}}{a}.$$

16.2 Effective Geometric Projection

The 00-component of the bridge metric equation yields the effective homogeneous projection

$$3F(\Phi)\left(H^2 + \frac{k}{a^2}\right) = \kappa\rho_\psi + \frac{\kappa}{2}Z(\Phi)\dot{\Phi}^2 + \kappa V(\Phi) - 3H\dot{F}(\Phi) + \kappa\rho_{\mathcal{A}},$$

where

$$\rho_{\mathcal{A}} := \gamma\,\mathcal{P}(\mathcal{A})\big|_{\text{hom}}.$$

Remark 45. *This equation is an effective geometric projection of the bridge system in a homogeneous regime. It is not imposed as the primitive cosmological law of the corpus.*

16.3 Homogeneous Coherence Evolution

The coherence equation reduces to

$$Z(\Phi)\left(\ddot{\Phi}+3H\dot{\Phi}\right)+\frac{1}{2}Z'(\Phi)\dot{\Phi}^{\,2}+V'(\Phi)-\frac{3}{\kappa}F'(\Phi)\left(\dot{H}+2H^2+\frac{k}{a^2}\right)+\frac{\partial\mathcal{L}_{\mathrm{m}}}{\partial\Phi}-\gamma\frac{\partial\mathcal{P}(\mathcal{A})}{\partial\Phi}=0.$$

16.4 Compatibility with Darke Theory Expansion Logic

The bridge theorem does not identify

$$H(t)=H_{\mathrm{rec}}(t)+H_{\mathrm{flow}}(t)$$

with the effective homogeneous projection above. Instead, the compatibility claim is:

> Whenever the Darke Theory cosmology enters a post-threshold regime admitting an effective homogeneous Lorentzian description, the geometric projection of that regime is governed by the bridge equations above. This does not require that the full cosmological architecture of Darke Theory be reduced to the standard Friedmann framework.

17 Final Bridge Theorem

Theorem 46 (Relativistic Closure–Curvature Bridge Theorem)**.** *Let M carry fields $(g_{\mu\nu},\Phi,\psi)$ satisfying:*

1. *post-threshold closure:*
$$\Phi>\Phi_{\mathrm{crit}};$$
2. *geometric activation:*
$$F(\Phi)>0;$$
3. *positive kinetic structure:*
$$Z(\Phi)>0,\qquad K(\Phi)>0;$$
4. *stationarity of the full bridge action:*
$$\delta S[g,\Phi,\psi]=0;$$
5. *hyperbolicity of the coherence principal symbol;*
6. *Lorentzian signature condition: the principal quadratic form has exactly one negative eigenvalue class;*
7. *finite admissibility:*
$$\mathcal{A}[\Phi,\psi;g]\geq 0.$$

Then:

(a) *the post-threshold regime admits an effective Lorentzian geometry;*

(b) *curvature is sourced by coherence gradients, matter sectors, and admissibility stress through*
$$F(\Phi)G_{\mu\nu}=\kappa T^{\mathrm{eff}}_{\mu\nu};$$

(c) *finite-energy nontrivial topological sectors of ψ act as matter-energy concentrations;*

(d) *physical propagation lies in the hyperbolic cone filtered by closure-balance admissibility;*

(e) *the coupled system is covariantly consistent on-shell;*

(f) *the post-threshold vacuum is linearly stable under the stated positivity conditions;*

(g) *whenever a homogeneous Lorentzian geometric sector is admitted, the bridge theory possesses the effective homogeneous closure-cosmology reduction stated above;*

(h) *the post-threshold regime therefore supports a concrete relativistic closure–curvature dynamics.*

Proof. Part (a) follows from Theorem 39. Part (b) is Theorem 34 in effective form. Part (c) follows from Theorem 40. Part (d) follows from Theorem 38. Part (e) follows from Theorem 42. Part (f) follows from Theorem 44. Part (g) is the homogeneous reduction established in the preceding section. Therefore the coupled system provides a coherent, admissible, relativistic bridge from post-threshold closure to curvature and matter dynamics. □

Interpretive Summary

The theorem chain established in this document activates geometry only in the post-threshold closure regime, defines a real admissibility functional with a smooth inadmissibility penalty, makes matter concrete through a closure-coupled complex scalar sector with finite-energy topological charge, derives a causal cone from hyperbolicity and refines physical propagation by closure balance, proves that the post-threshold regime is Lorentzian under explicit signature conditions, proves that stable matter sectors source curvature, proves on-shell gravitational consistency, exhibits both weak-field and homogeneous closure-cosmology reductions, and proves linear stability of the post-threshold vacuum under explicit positivity assumptions.